

RAZ EINI

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PROFILE

Software Engineering student (Atuda) targeting cyber and intelligence units, with hands-on experience in defensive security, network traffic analysis, and cryptographic implementations. Demonstrated ability to build production-grade tools, including a multi-threaded NIDS with active defense and Grafana observability, alongside a cryptographically secure CLI vault with automated CI/CD pipelines. Proficient in Linux environments and driven to apply low-level security concepts to high-impact cyber operations.

SECURITY & TECHNICAL SKILLS

- **Security & Observability:** Deep Packet Inspection (DPI), NIDS/IPS logic, Active Threat Mitigation (Dynamic IP Isolation), Grafana & Loki (Dashboards as Code), Promtail.
- **Networking & Protocols:** Packet Capture & Analysis (Scapy, Raw Sockets), OSI/TCP-IP Stack, Dynamic Memory Garbage Collection, Thread-Safe Concurrency (threading, queue.Queue).
- **Cryptography & Systems:** Fernet (AES-128-CBC & HMAC), PBKDF2-HMAC-SHA256, Salting & Hashing, Key Derivation Functions (KDF), Linux Systems & CLI Administration, Log Analysis (OverTheWire Bandit).
- **Languages & Infrastructure:** Python, Java, Kotlin, Bash, Git & GitHub Actions (CI/CD), Docker & Docker Compose, Android (Jetpack Compose, MVVM), Firebase.

PROJECTS

NetGuard — Full-Stack NIDS & Observability Engine | *Python, Scapy, Docker Compose, Grafana, Loki, Promtail*

- Architected a full-stack Network Intrusion Detection System (NIDS) featuring multi-threaded Deep Packet Inspection (DPI) and dynamic active defense (IP isolation) mechanisms.
- Engineered a high-performance producer-consumer pipeline using queue.Queue and thread-safe locking to handle real-time traffic analysis without packet loss or memory leaks, including a background garbage collector.
- Integrated an automated observability stack (Grafana, Loki, Promtail) using "Dashboards as Code" to visualize security events, threat timelines, and anomalous IP behavior in real time.

Secure CLI Password Manager | *Python, Fernet (AES-128-CBC + HMAC), CI/CD*

- Architected a modular password manager using a clean src-layout structure, ensuring robust separation between the cryptographic core and CLI interfaces.
- Implemented high-security standards including PBKDF2-HMAC-SHA256 key derivation (390k iterations) and Fernet (AES-128-CBC + HMAC) authenticated encryption to ensure data integrity and confidentiality.
- Modernized the deployment lifecycle with a full CI/CD pipeline and automated pytest suites on GitHub Actions, ensuring cross-version code quality and production-ready robustness.

E-Commerce Shop App | *Android, Java, Firebase*

- Developed a scalable shopping application with real-time Firebase database synchronization, multi-device cart consistency, and role-based admin inventory management.

Smart TaskFlow | *Kotlin, Jetpack Compose, MVVM*

- Engineered a task management application using Jetpack Compose and MVVM architecture, ensuring business logic decoupling and clean state management.

EDUCATION

B.Sc. Software Engineering, Afeka College of Engineering - Present | Expected Graduation: 2029

Academic Reserve (Atuda) "Amirim" Program — pre-service technological track

Makif Tet High School | 2022 – 2025

High School Diploma (GPA: 91): Majors in Software Engineering & Cyber (10 units), Physics, Math, and English (5 units).

Key Achievements: Grade 100 in Systems Design & Programming; Software Technician Certification.

ADDITIONAL

- **Languages:** Hebrew (native), English (high level)
- **Interests:** Chess (Peak 2200 ELO), speedcubing